

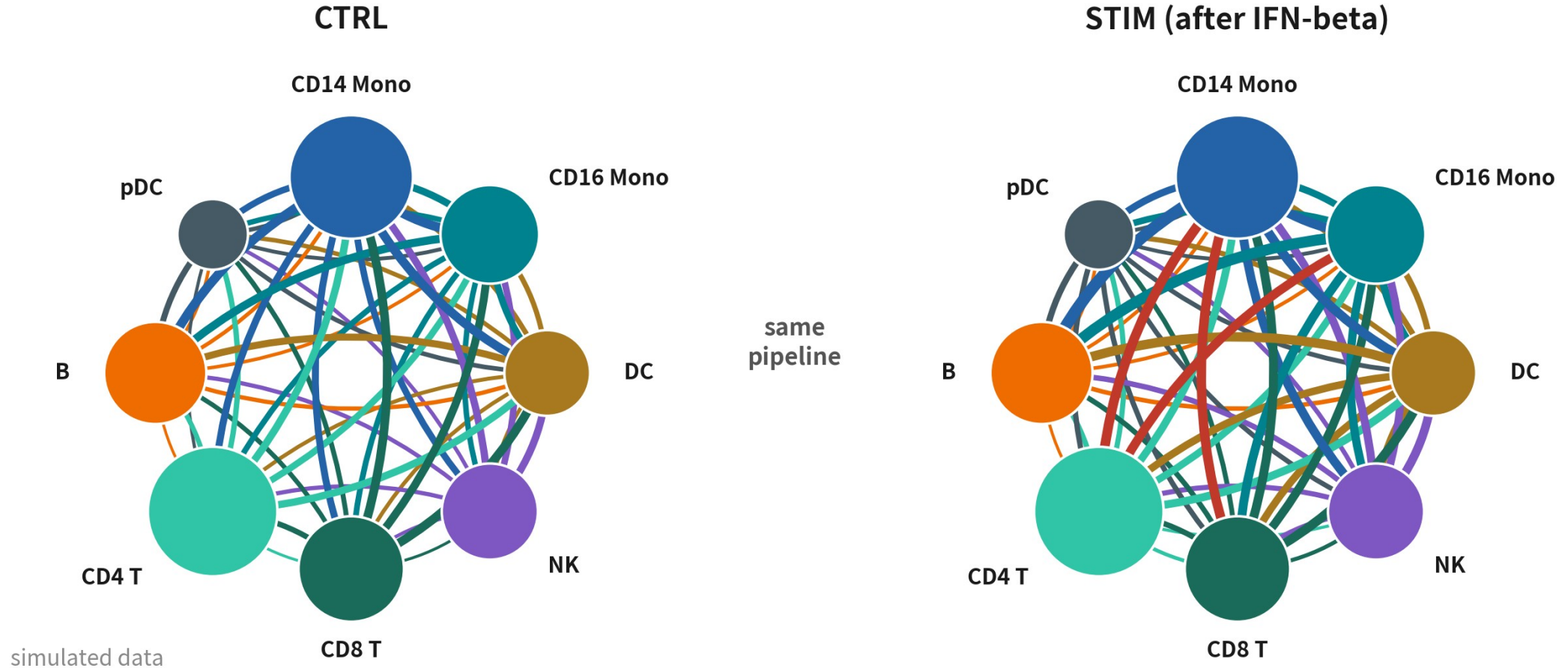
B14 Series B

Cell-cell communication: ligand-receptor pairs with CellChat

Infer the network, compare conditions — and know exactly where the inference ends.

About 40 min | Prerequisites: B10-B12 | Data: ifnb (from the B11 object)

Two circle plots, one thickened edge



The monocyte → T edge tripled in width. Discovery — or artifact?

After stimulation the monocyte → T edge tripled. Discovery, or artifact?

WHERE THIS EPISODE STARTS

**Communication analysis is inference,
not observation.**

No step ever watches two cells talk. We keep this sentence in hand for the whole episode.

What you will learn

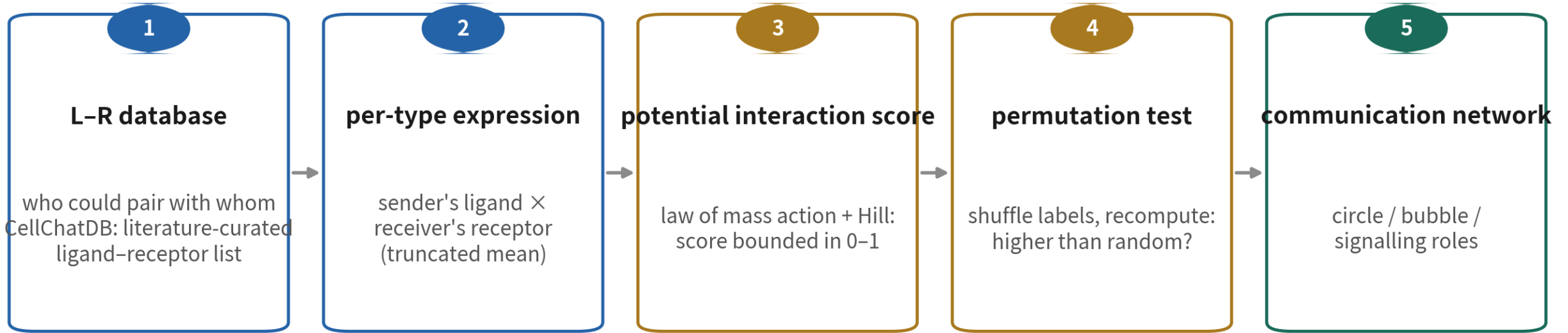
- 1 State the inference chain and its three fatal assumptions
- 2 Run the full CellChat pipeline: circle plots, bubble plots, signalling roles
- 3 Compare CTRL vs STIM networks and pinpoint condition-specific axes

01

How communication gets inferred

Chain, assumptions, database — foundations before code

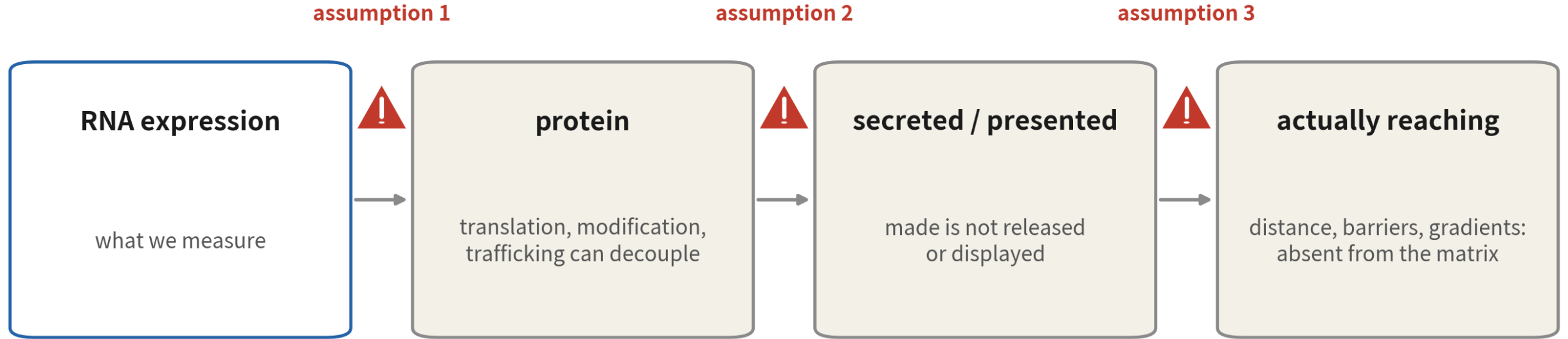
The five-step inference chain



Every step is inference — nothing here ever observes two cells touching

Database $\times$ expression $\rightarrow$ score $\rightarrow$ permutation test $\rightarrow$ network. Every step is inference

Three fatal assumptions



Three arrows = three unverified assumptions. The tool still hands you a pretty network.

So the claim is 'predicted communication' — validation needs imaging or functional assays

RNA is not protein, not secretion, not contact — the tool validates none of these arrows

CellChatDB: where the list comes from

Three categories in CellChatDB.human

Secreted Signaling



~60%

ECM-Receptor (matrix)



~22%

Cell-Cell Contact



~18%

proportions illustrative; cytokines and chemokines are mostly secreted

One row of the database (one interaction)

CXCL10 - CXCR3

simple ligand-receptor pair

IL12A+IL12B - IL12RB1+IL12RB2

both sides are complexes

TGFB1 - TGFB1+TGFB2

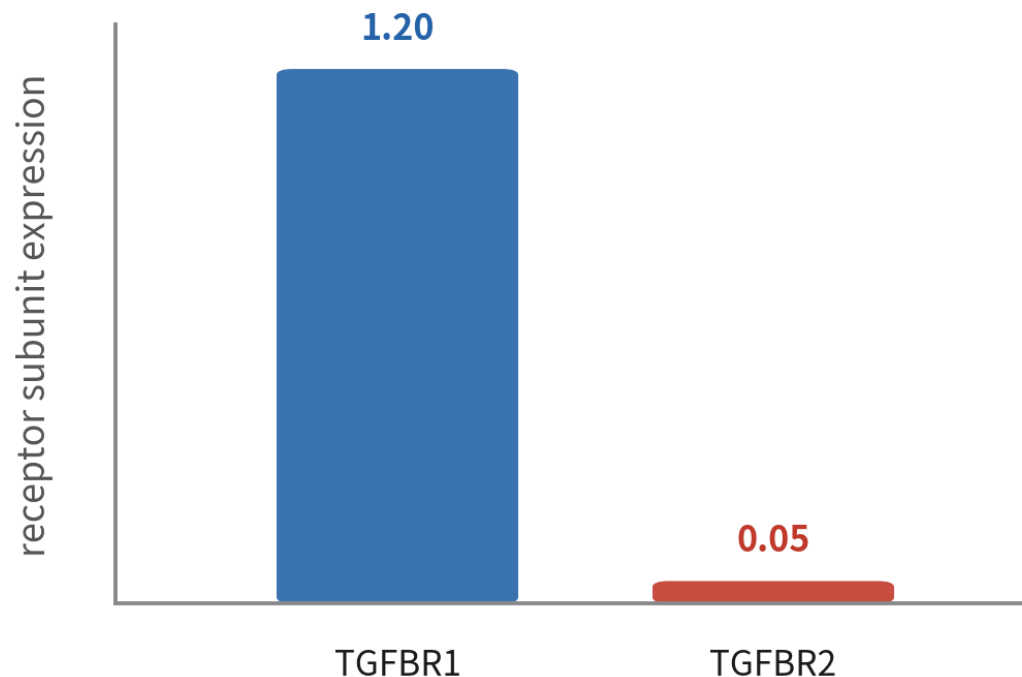
two-subunit receptor complex

multi-subunit: missing any subunit should kill the interaction

- a curated ligand-receptor database in three categories
- secreted signalling dominates; contact and ECM take the rest
- interactions can involve complexes: multi-subunit is modelled explicitly
- anything absent from the database can never appear in your results

Multi-subunit: why the geometric mean

One cell type's two subunits



arithmetic mean (naive)



$$(1.2 + 0.05) / 2 = 0.62$$

high score despite a missing subunit — false positive

geometric mean (CellChat)



$$\sqrt{1.2 \times 0.05} = 0.24$$

any subunit near zero drives the score to zero

If any subunit approaches zero the complex's score approaches zero — by design

CORE IDEA

CellChat hands you a ranked list of hypotheses about who could be talking — not evidence that they are.

A hypothesis list is valuable — it tells you what to validate next. It is not a conclusion.

02

Data prep and createCellChat

The B11 integrated object + B10-style labels

Load the object, set the labels

```
library(Seurat)
library(CellChat)
set.seed(1234)

ifnb <- readRDS("output/ifnb_b11_integrated.rds")
ifnb <- JoinLayers(ifnb) # 通訊分析用 RNA 原值
Idents(ifnb) <- "seurat_annotations" # 可靠的型別標籤 (B10)
sort(table(Idents(ifnb)), decreasing = TRUE)[1:4]
```

CD14 Mono	CD4 Naïve T	CD4 Memory T	CD16 Mono
4362	2504	1762	1044

Label quality equals conclusion quality — without B10 this episode is castles in the air

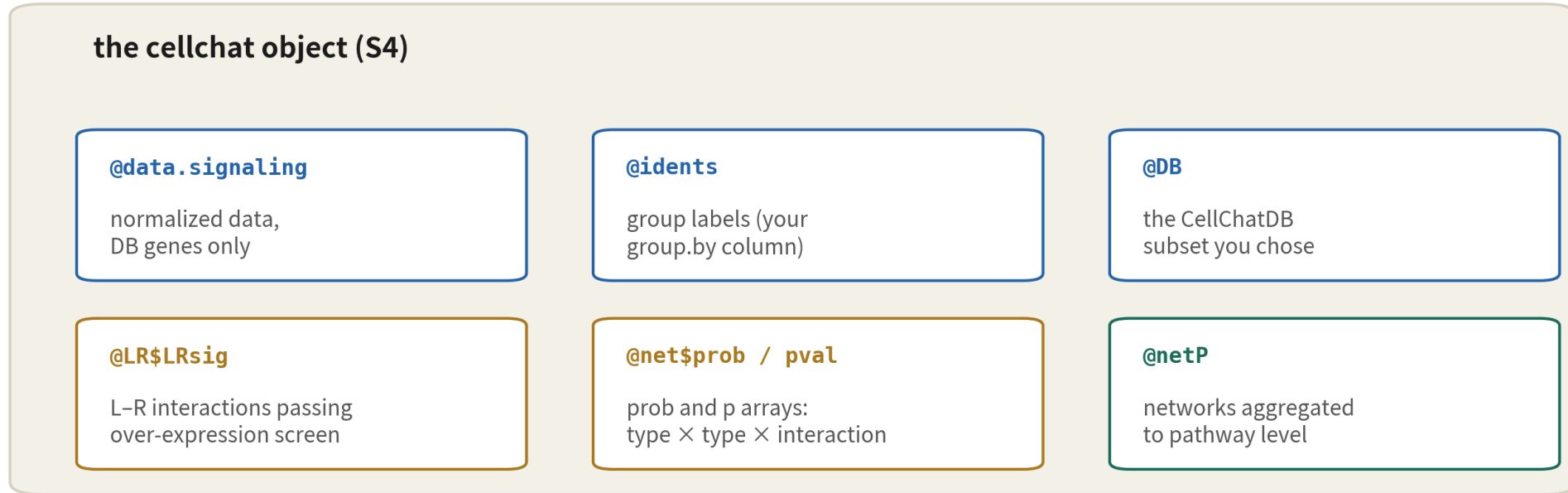
Subset CTRL, create the CellChat object

```
ctrl<-subset(ifnb,subset= stim == "CTRL")  
cc.ctrl<-createCellChat(object= ctrl,  
                        group.by = "ident", assay = "RNA")  
cc.ctrl@DB <- CellChatDB.human      # 掛上人類 L-R 資料庫
```

Create a CellChat object from a Seurat object
The 'data' slot in the default assay is used ...
The cell groups used for CellChat analysis are
B, B Activated, CD 14 Mono, CD 16 Mono, ...

One object per condition — comparison happens later, in mergeCellChat

Anatomy of the cellchat object



Top row: filled at creation. Bottom row: filled step by step by the inference pipeline

- top row: filled at creation (data, labels, database)
- bottom row: filled step by step by the pipeline
- @net\$prob is the core output: a type × type × interaction array

What you have at this point

One CTRL object



6,548 cells, 13 type
labels, log-normalized
data

One L-R list



CellChatDB.human,
with multi-subunit
complexes

No scores yet



the four inference calls
come next, in one pass

03

The core: from over-expression to probability

identifyOverExpressedGenes → computeCommunProb

Screen out the impossible interactions

```
cc.ctrl <- subsetData(cc.ctrl) # 只留資料庫基因，必跑  
cc.ctrl <- identifyOverExpressedGenes(cc.ctrl)  
cc.ctrl <- identifyOverExpressedInteractions(cc.ctrl)  
nrow(cc.ctrl@LR$LRsig) # 進入計分的互動數
```

```
[1] 1170
```

Ask 'which pairs are even plausible' before spending compute — the number varies by dataset

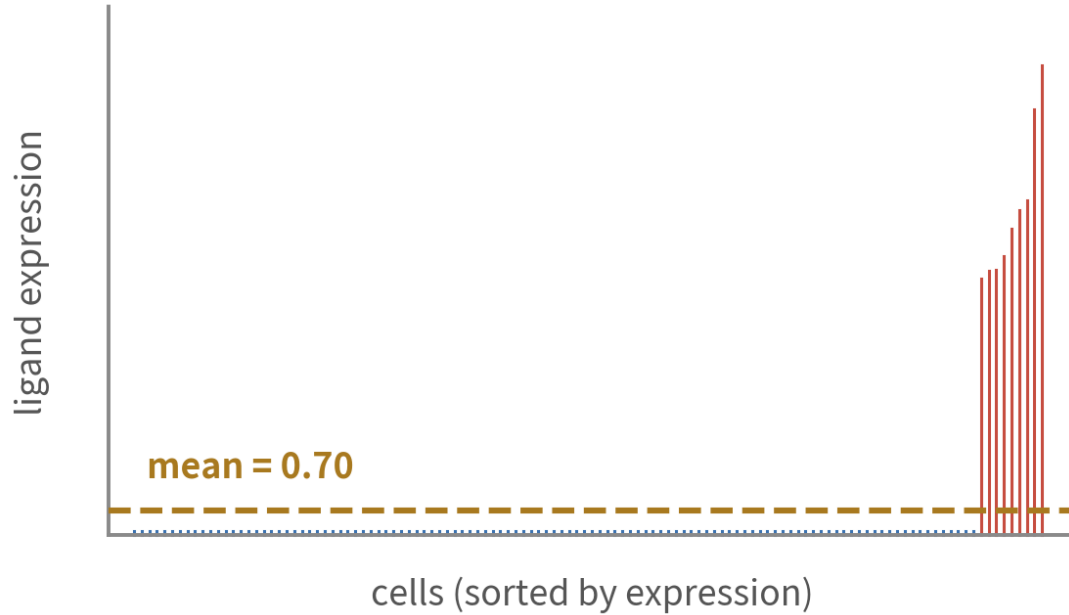
computeCommunProb: the line of the episode

```
cc.ctrl <- computeCommunProb(cc.ctrl, type = "trim",  
                             population.size = FALSE)  
cc.ctrl <- filterCommunication(cc.ctrl, min.cells = 10)
```

type sets how per-type expression is summarised; population.size decides whether abundance counts

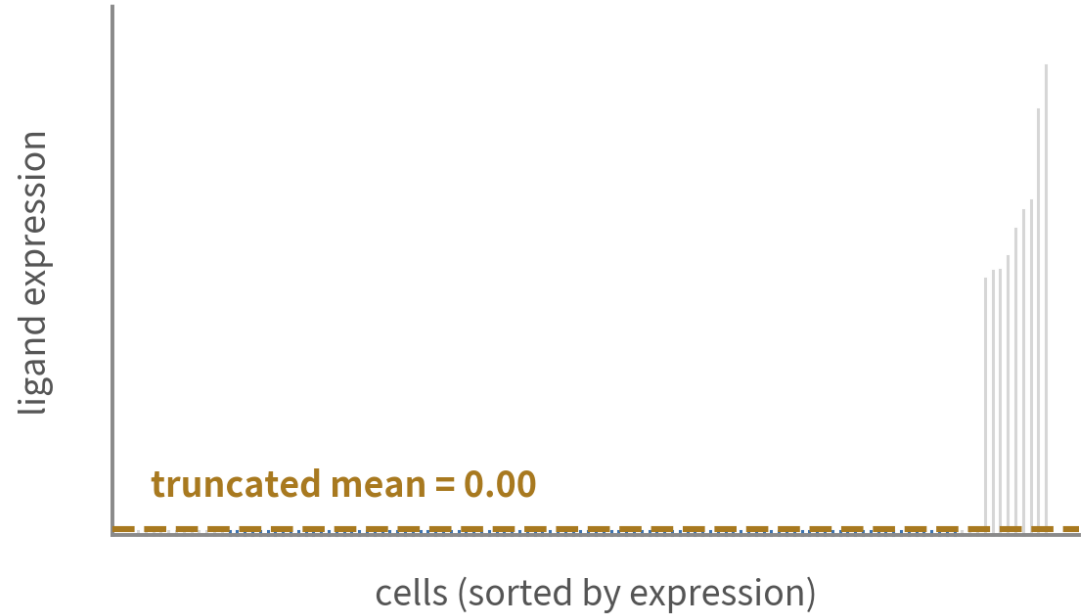
Trimming: body armour for the type mean

arithmetic mean: hijacked by 9 hot cells



a few cells' fireworks get averaged into 'this type expresses it'

10% truncated mean: drop 10% each end, then average



expression too rare to survive the trim → score zero: the pair does not count

simulated data

Default triMean needs roughly 25% of cells expressing; truncatedMean with trim = 0.1 relaxes it to 10%

What the permutation test does

Permutation test for CXCL10 → CXCR3 (CD14 Mono → CD8 T)



simulated data

The p-value answers 'higher than random?', not 'how strong biologically' — trap 2 returns to this

Shuffle labels, recompute 100 times — a small p means the observed beats random

Aggregate to pathways and the overall net

```
cc.ctrl <- computeCommunProbPathway(cc.ctrl)  
cc.ctrl <- aggregateNet(cc.ctrl)  
groupSize <- as.numeric(table(cc.ctrl@idents))
```

L-R level → pathway level → type × type totals: all three resolutions live in the object

04

The visualisation trio

circle → bubble → signalling roles, coarse to fine

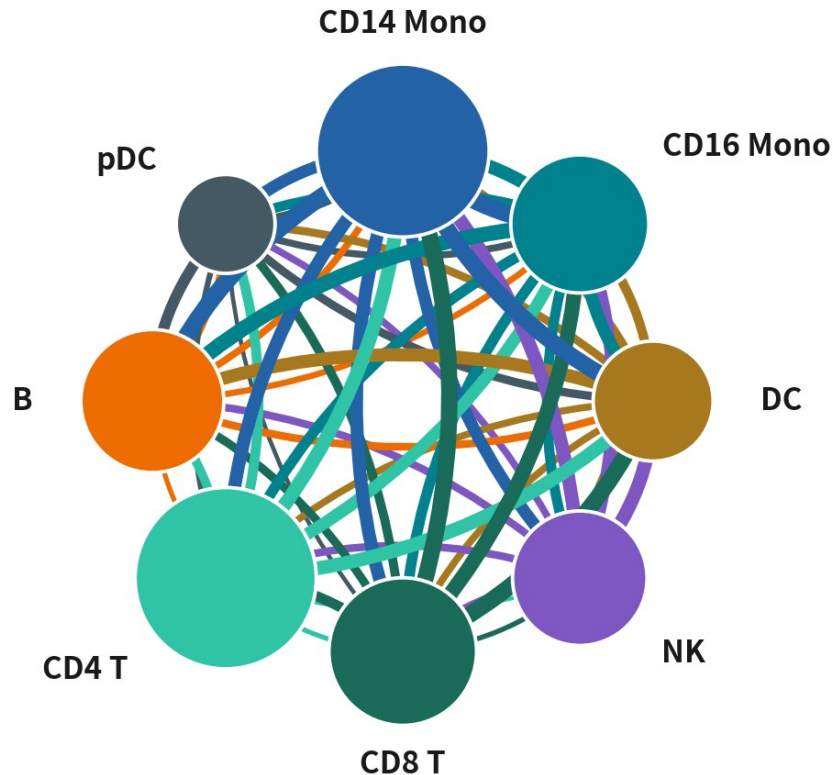
Piece one: netVisual_circle

```
par(mfrow = c(1,2), xpd = TRUE)
netVisual_circle(cc.ctrl@net$count, vertex.weight = groupSize,
  weight.scale = TRUE,
  title.name = "Number of interactions")
netVisual_circle(cc.ctrl@net$weight, vertex.weight = groupSize,
  weight.scale = TRUE,
  title.name = "Interaction strength")
```

count and weight are different plots: how many edges vs how strong — keep them apart

How to read the circle plot

The aggregated network (CTRL)



simulated data

- node size = number of cells of that type
- edge colour = the sender (ligand side)
- edge width = interaction count or strength (count and weight are two separate plots)
- arrowhead = ligand → receptor direction
- read globally first: who shouts, who listens — then zoom into bubbles for the L-R pairs

Find the loudspeakers and the listeners first — do not fixate on one edge yet

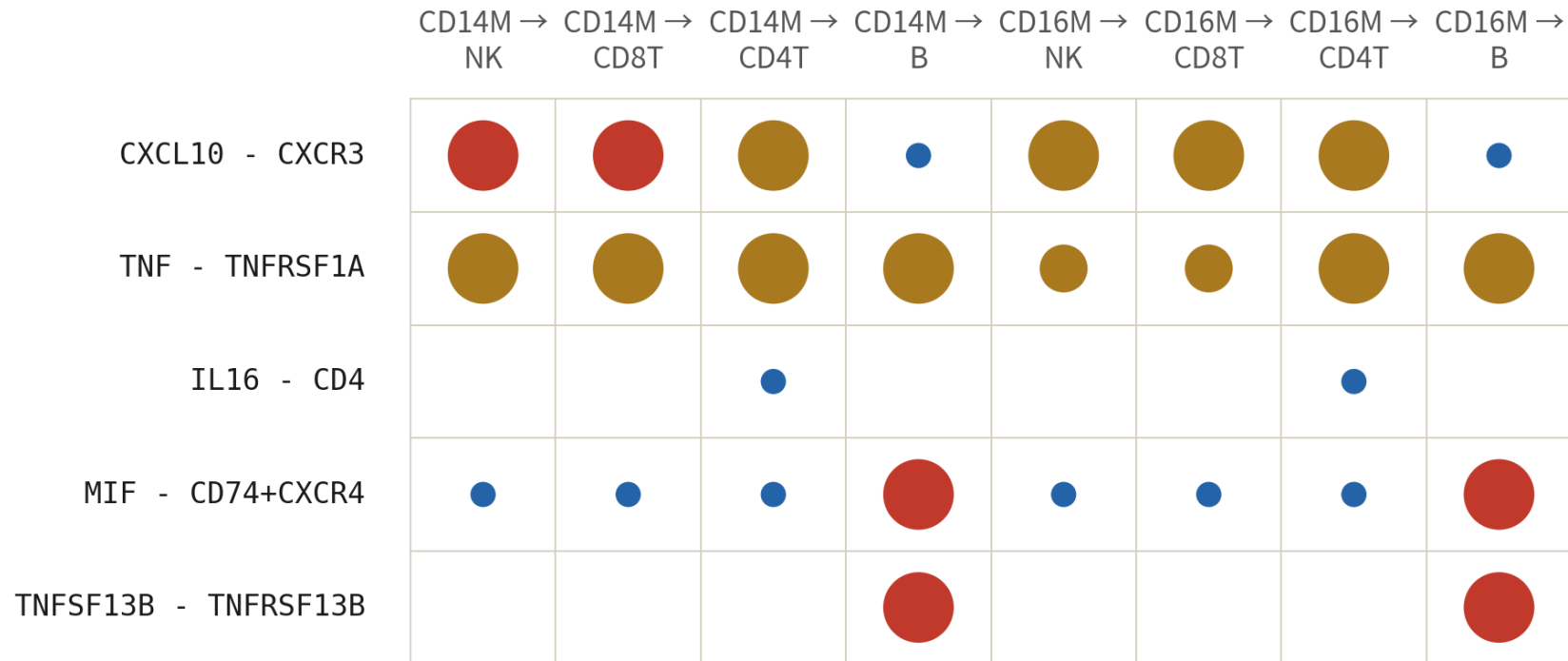
Piece two: netVisual_bubble

```
netVisual_bubble(cc.ctrl,  
    sources.use = c("CD 14 Mono", "CD 16 Mono"),  
    targets.use = c("NK", "CD 8 T",  
        "CD 4 Naïve T", "B"),  
    remove.isolate = TRUE)
```

Circle gives direction, bubble gives names: which L-R pairs carry that edge

How to read the bubble plot

netVisual_bubble: monocyte → lymphocyte L-R detail



how to read

- one row = one L-R pair
- one column = one sender → receiver
- dot size = permutation p
(big = $p \leq 0.01$)
- dot colour = communication prob.
- blank = below threshold, not zero

simulated data

Dot size is the p-value, colour the probability — read both; trap two lurks here

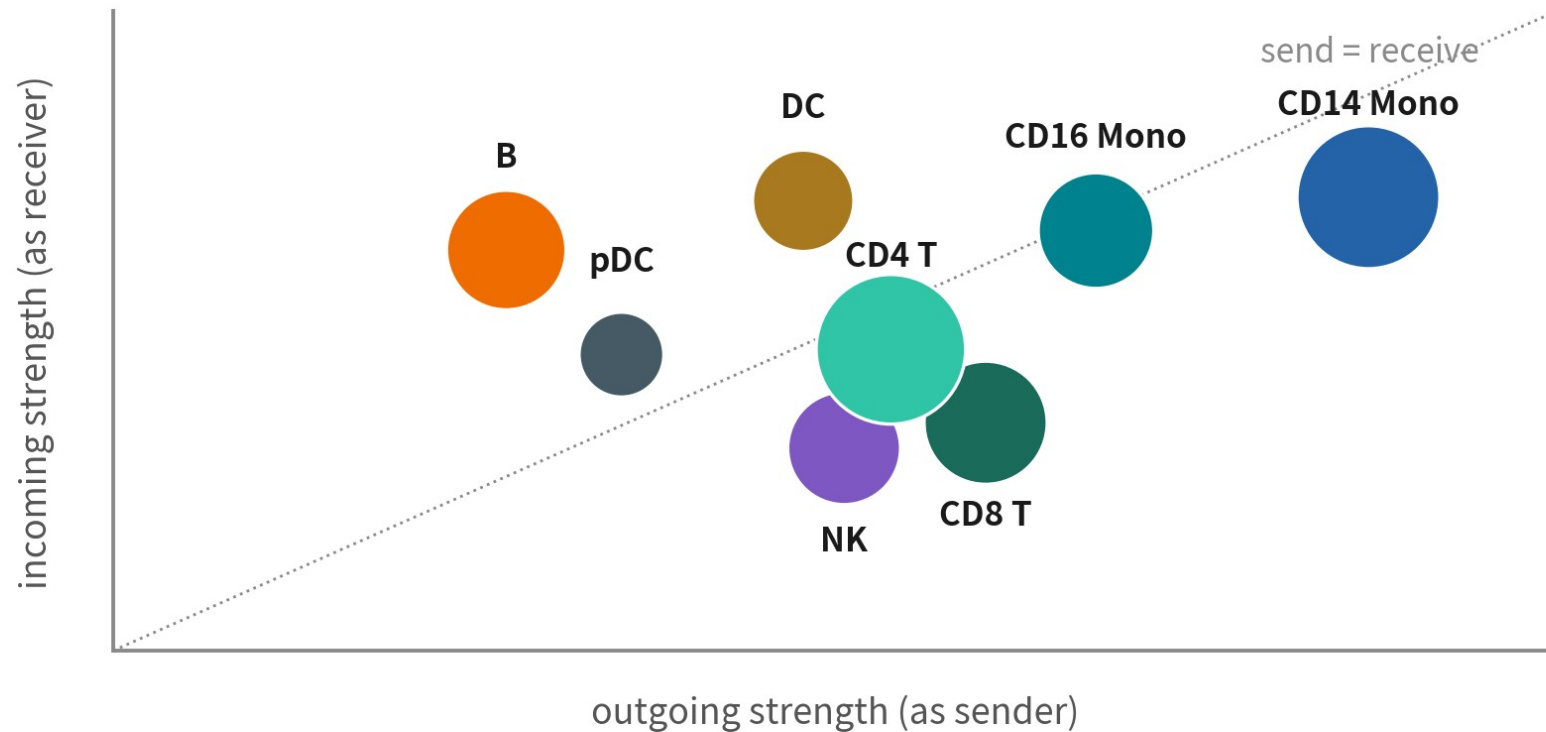
Piece three: signalling-role analysis

```
cc.ctrl<-netAnalysis_computeCentrality(cc.ctrl,  
                                       slot.name = "netP")  
netAnalysis_signalingRole_scatter(cc.ctrl)
```

Summarises each type's outgoing and incoming strength as network centrality

In / out: who sends, who receives

Signalling role of each type: who sends, who receives



read it

bottom-right:
main senders
(monocytes)

top-left:
main receivers
(B, pDC)

on diagonal:
two-way hubs

simulated data

Monocytes are the dominant senders here; B cells and pDC mostly listen

What each piece answers

circle: the whole



who signals whom,
how strongly — the
global glance

bubble: the detail



which L-R pairs carry a
given edge

roles: the division



is each type a sender, a
receiver, or a hub

05

CTRL vs STIM: comparing two networks

Separate objects → mergeCellChat → rankNet and the differential net

Same pipeline for STIM, then merge

```
run_cellchat <- function(obj) {  
  cc <- createCellChat(obj, group.by = "ident", assay = "RNA")  
  cc@DB <- CellChatDB.human  
  cc <- subsetData(cc) |> identifyOverExpressedGenes() |>  
    identifyOverExpressedInteractions() |>  
    computeCommunProb() |>  
    filterCommunication(min.cells = 10) |>  
    computeCommunProbPathway() |> aggregateNet()  
  netAnalysis_computeCentrality(cc, slot.name = "netP")  
}  
cc.stim <- run_cellchat(subset(ifnb, subset = stim == "STIM"))  
cc <- mergeCellChat(list(CTRL = cc.ctrl, STIM = cc.stim),  
  add.names = c("CTRL", "STIM"))
```

Wrapping it in a function guarantees identical parameters — the only way a comparison means anything

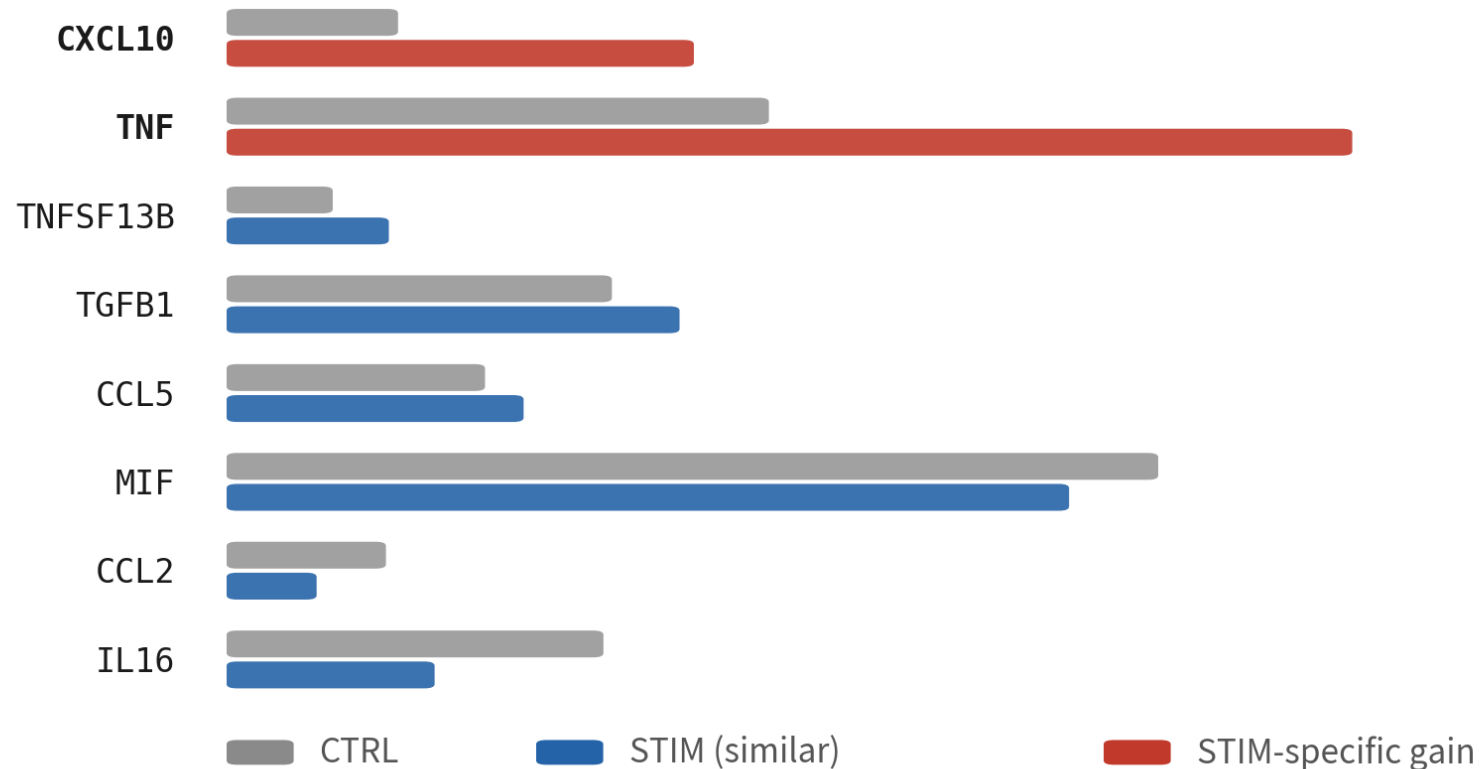
Compare totals and information flow

```
compareInteractions(cc, group = c(1, 2)) # 總互動數  
rankNet(cc, mode = "comparison", stacked = TRUE,  
        do.stat = TRUE) # 每條 pathway 的資訊流
```

rankNet's 'information flow' is the summed communication probability of a pathway

rankNet: which pathways got rewritten

rankNet: information flow (summed prob) per pathway



CXCL10 and TNF surge after stimulation — downstream axes of the interferon response

CCL2 and IL16 weaken: the flow is rewritten, not uniformly boosted

simulated data

CXCL10 and TNF surge, CCL2 and IL16 fade — echoing the B12 DE and B13 GSEA results

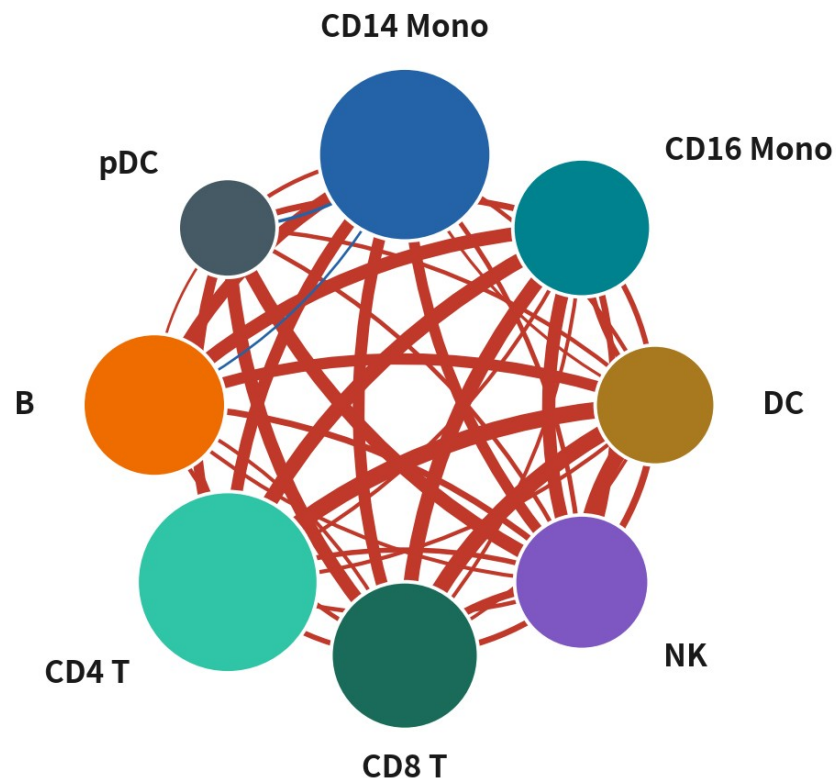
The differential network in one plot

```
par(mfrow = c(1, 1))  
netVisual_diffInteraction(cc, weight.scale = TRUE,  
                           measure = "weight")  
# 紅 = STM 比 CTRL 強; 藍 = 變弱
```

Subtract the two nets and plot only the difference — changes pop out better than side-by-side

Reading the differential network

Differential network: STIM – CTRL



simulated data

■ up in STIM

■ down in STIM

- red: monocyte-derived signals broadly gain (CXCL10, TNF, TNFSF13B)

- blue: axes like CCL2 and IL16 weaken after stimulation

- edge width = |difference|

- this answers 'which axes changed'; whether it is biology needs effect sizes + validation

Monocyte-derived signals broadly gain — biologically sensible under IFN-beta

FOR YOUR METHODS SECTION

"CellChat analysis suggests enhanced monocyte-derived CXCL signalling after stimulation ($p < 0.01$), pending experimental validation."

The verb is 'suggests' or 'predicts', never 'demonstrates'. Not modesty — accuracy.

The ecosystem: two other major tools

Same database × expression family — each adds one extra move

CellPhoneDB

strict multi-subunit statistics

- pioneered the all-subunits-or-nothing rule
- permutation framework akin to CellChat's
- Python ecosystem; Efremova 2020 Nat Protoc

NicheNet

asks one more question: did downstream respond

- links ligands to target-gene changes in receivers
- answers 'which ligand explains these DE genes'
- Browaeys 2020 Nat Methods

06

Where people get hurt

Three traps — each demonstrated for real

Trap 1: scoring tiny clusters anyway

What it is

Shrinking (or skipping) filterCommunication's min.cells so a 20-cell group still scores.



Why it happens

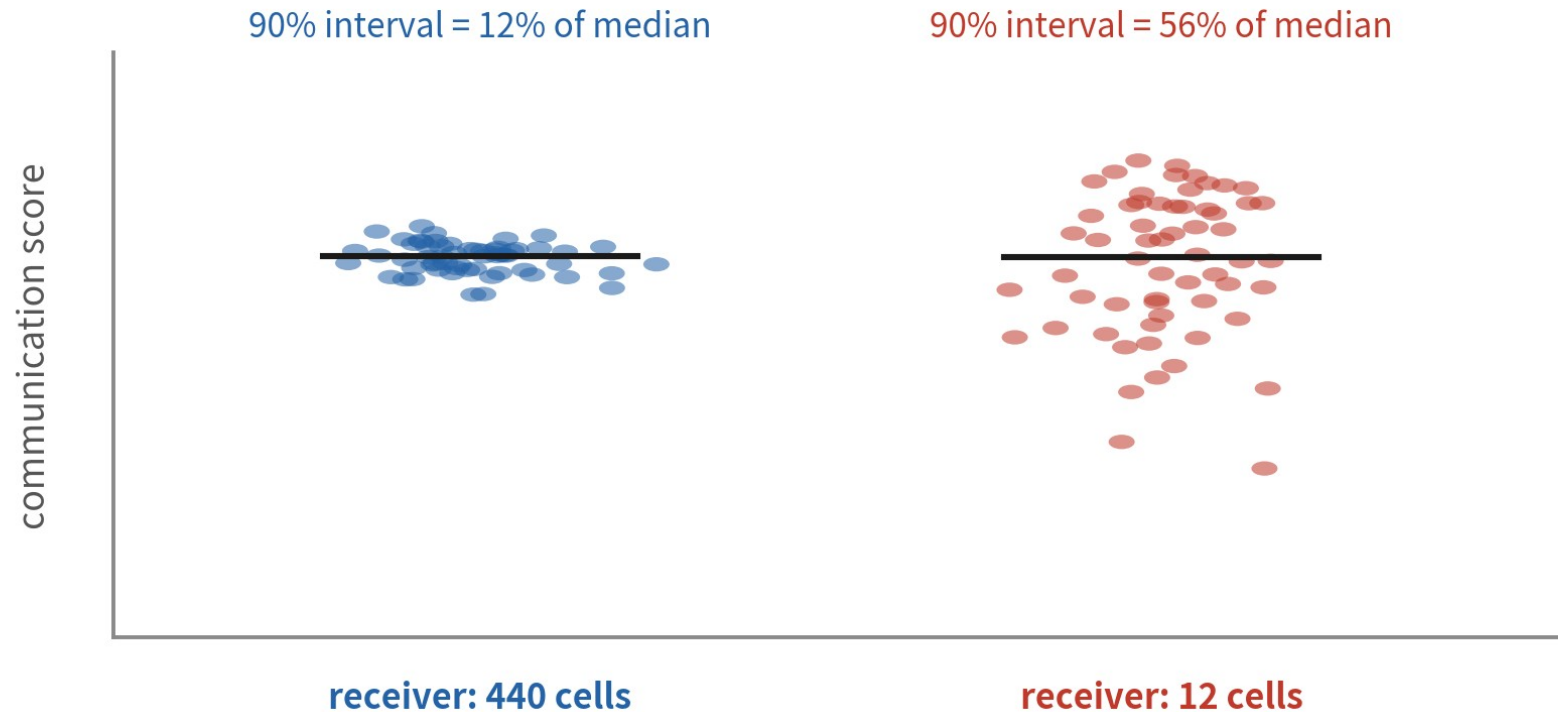
Tiny clusters are often the rare populations you care about most — hard to filter away.

What it costs you

The truncated mean hangs on 2-3 cells; one resample flips the result. Those 'strong interactions' are dice rolls.

Demonstrated: tiny-cluster scores are dice

Same interaction, 60 resamples: tiny clusters roll dice



a truncated mean over 12 cells is decided by 2–3 cells

remedy:
filterCommunication
(min.cells = 10) is a floor,
not a licence — distrust
tiny-cluster conclusions

simulated data

Same interaction, 60 resamples: with 12 cells the 90% interval exceeds half the median

Trap 2: reporting scores as effect sizes

What it is

Writing 'X→Y communication is significant ($p < 0.01$), so X strongly regulates Y'.



Why it happens

A tiny p feels like a big effect — intuition lies exactly here.

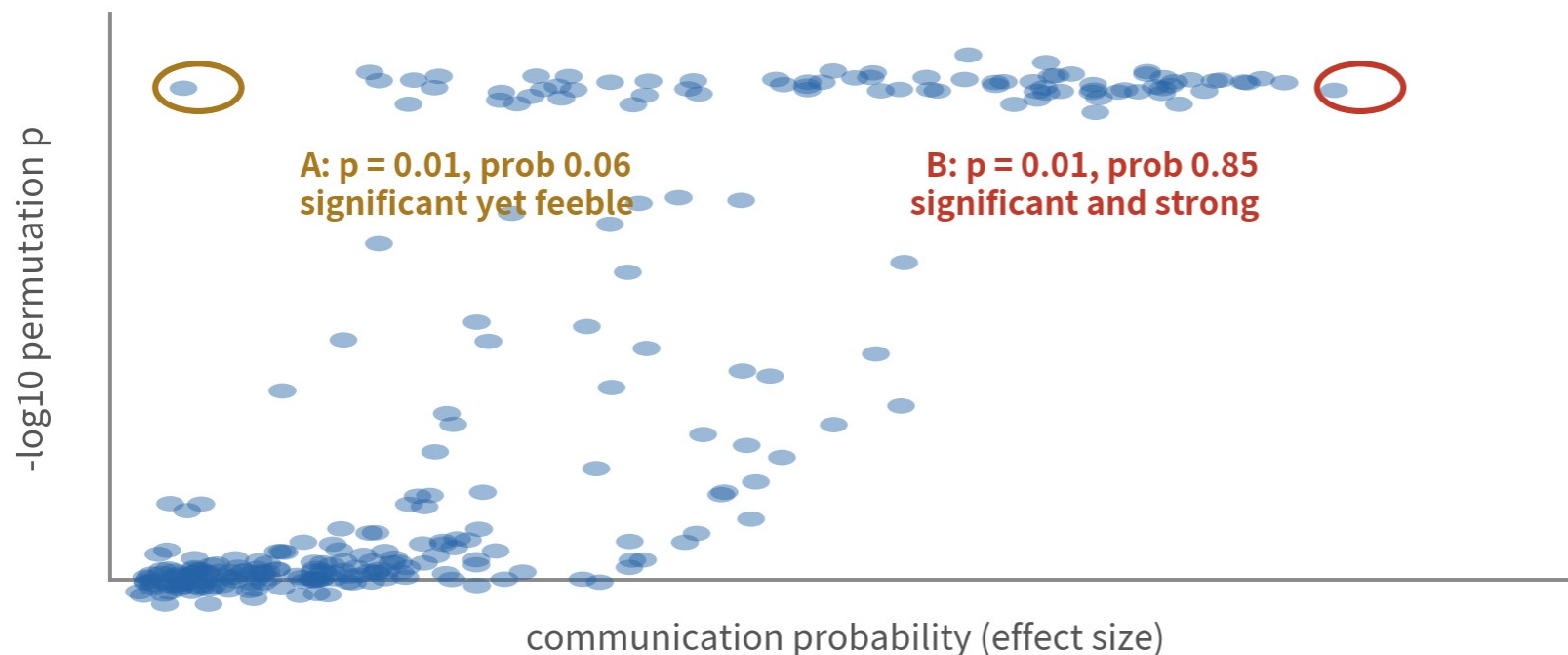
What it costs you

The permutation p only says 'beats shuffled labels'; with specific expression even feeble signals hit the p floor. Strength lives in the probability.

Same p, wildly different interactions

Same p-value, wildly different interactions

top band = p at its floor (1/100 with 100 permutations)



the permutation p only says
'higher than shuffled labels'

report in pairs:
probability (strength) + p

with specific expression,
even tiny scores go significant

simulated data

Always report in pairs: probability (strength) plus p-value (stability)

Trap 3: cluster-granularity mismatch

What it is

Running communication on an arbitrary resolution — 8 vs 14 clusters yield different headline conclusions.



Why it happens

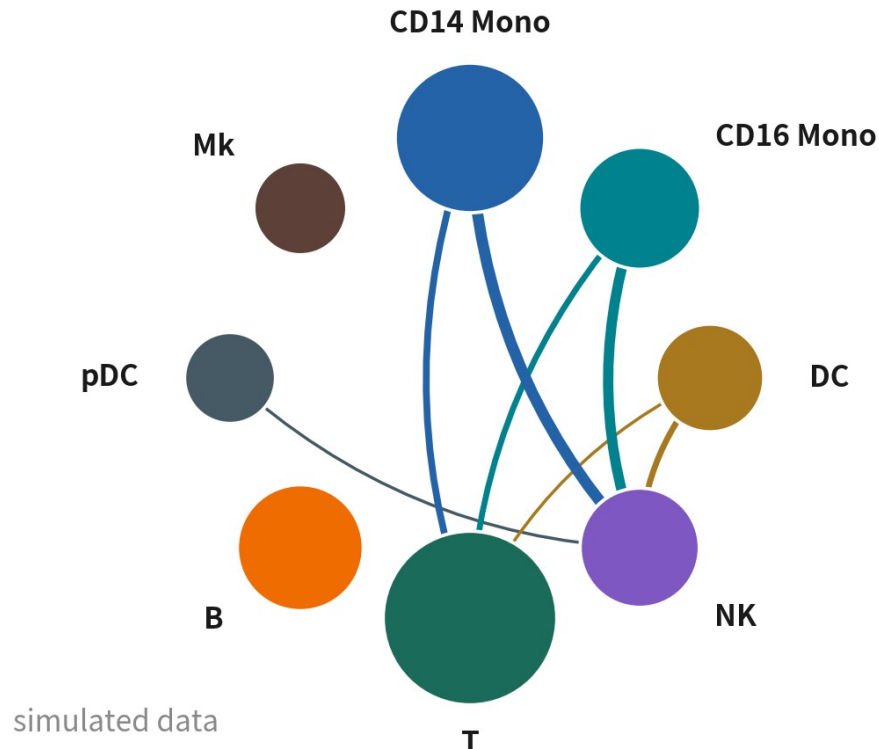
Scores are computed on type means; merging a receptor-specific subset into a big cluster dilutes it away.

What it costs you

Same data, two granularities: the top axis flips from monocyte → NK to monocyte → activated T. Granularity is part of the claim.

Demonstrated: 8 vs 14 clusters flip the story

8 clusters: all T cells merged

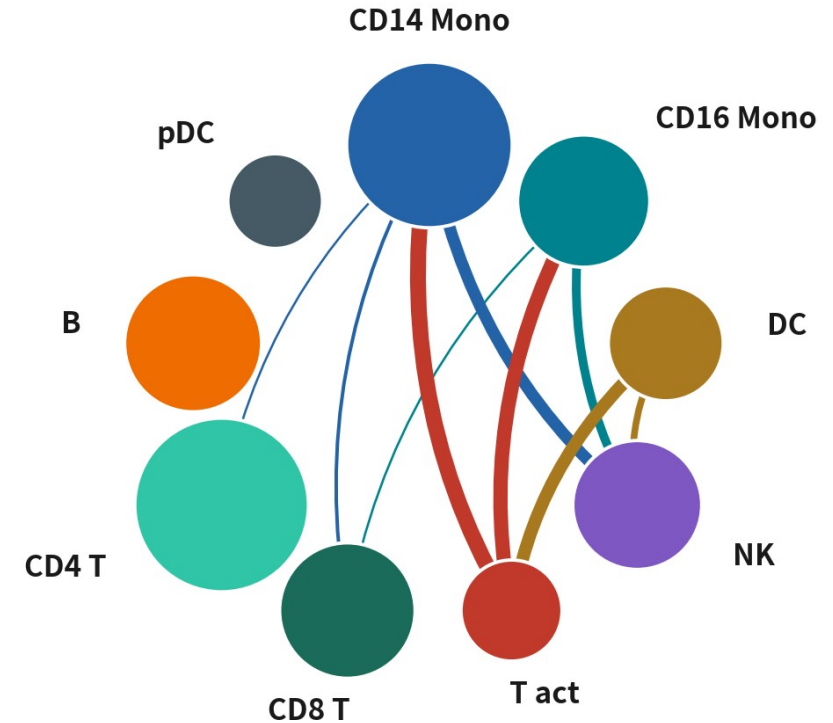


conclusion: 'monocytes signal mainly to NK'

CXCR3 lives on the 5% activated subset: merging dilutes it away and the axis vanishes

The receptor lives on the 5% activated subset: merge, dilute, and the axis vanishes

14 clusters: activated T split out (9 shown)



conclusion: 'the top axis is monocyte → activated T'

TRAPS, SUMMED UP

**Three gates before you write it up:
is the cluster big enough? the strength real?
the granularity defensible?**

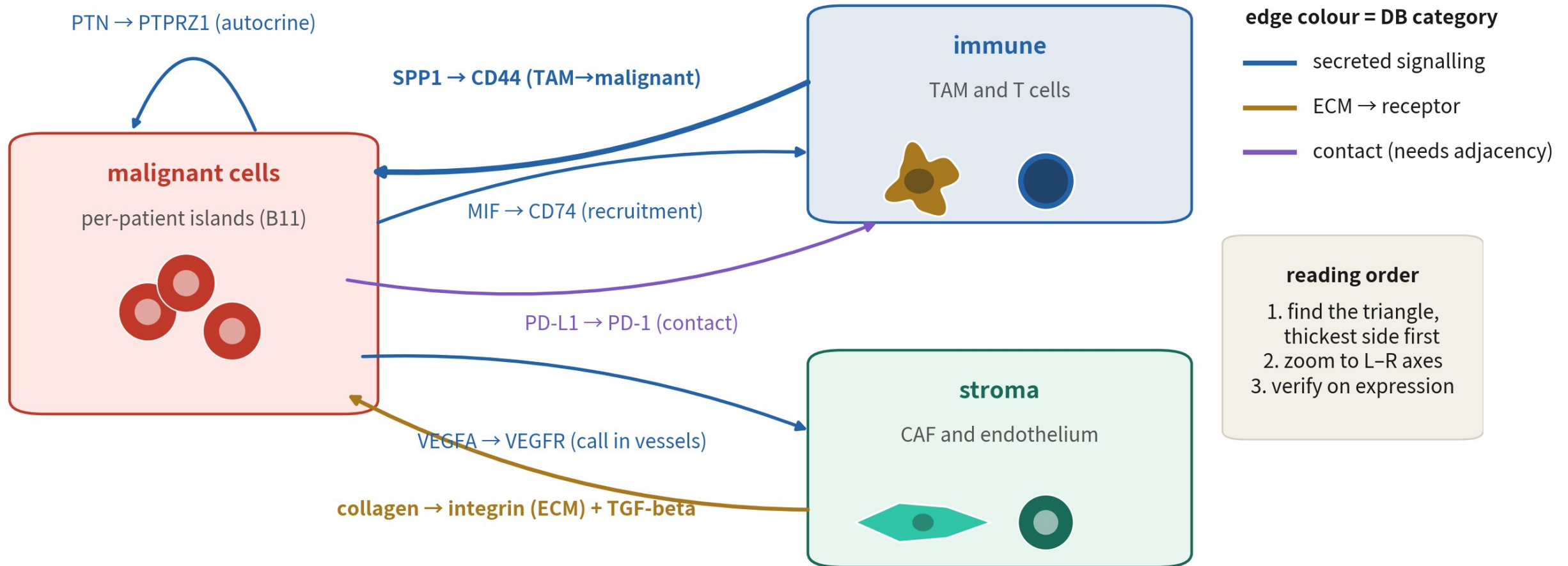
Even past all three gates it is still a prediction. Keep the verb at 'suggests'.

06+

Strategy Room: Complex Samples

The same CellChat, on a multi-patient tumour — how to read the map, and which decisions to remake

The TME triangle: tumour, immune, stroma



schematic; axes compiled from GBM / melanoma / breast-cancer literature — trust your own run

Tumour communication maps share one skeleton: find this triangle, ask which side is thickest and which axis carries it — then zoom in

Malignant L-R pairs deserve one extra layer of doubt

On top of the three assumptions, malignant cells add three more checks

CNV shifts expression



a ligand or receptor on an amplified segment reads high because of copy number. GBM often gains all of chr7 — EGFR sits on it; before trusting an EGFR axis, check the B16 CNV call

Per-patient communication



patient A's macrophages cannot talk to patient B's tumour cells — one object per patient per sample, with post-CNV malignant labels

Can the ligand even leave the cell



consensus databases mix in co-expression records; intracellular proteins do top ligand lists — check subcellular location yourself, no tool does it for you

Famous axes, tumour by tumour

Priors calibrate direction, they guarantee nothing — validate what appears, and absence is not an error

Tumour	Signature axis	Reading note
GBM	tumour-myeloid: TAM→malignant SPP1-CD44; malignant→TAM MIF-CD74; PTN autocrine	the thickest edge is almost always malignant-macrophage; faint T rows are expected — an immune-cold tumour
Melanoma	the PD-1 axis: tumour / myeloid PD-L1 → T-cell PD-1	a contact-type interaction — dissociated data only shows both ends expressed; adjacency needs spatial proof. The action is on the T-cell side
Breast cancer	the CAF axis: CAF→tumour collagen-integrin (ECM), TGF-beta	the stroma carries real weight; CAF is not one cell type — split the subsets before naming axes

Two-condition comparison: designing pre vs post

mergeCellChat is the tool from the main line — what changes in tumours is the design

Mixed run

Three fake differences

- one object over all patients: patient A's ligand meets patient B's receptor — a conversation that does not exist
- if patient composition shifts between timepoints, interaction totals track composition, not biology
- an axis 'lost after treatment' may just be a cell group dropping below min.cells and being removed wholesale

Paired per patient

Compare within patients, demand consistency across them

- one object per patient per timepoint, the pipeline wrapped in a function so parameters match exactly
- mergeCellChat within each patient: rankNet (do.stat) ranks information flow, the differential net shows direction
- only axes consistent across patients make the main figure; single-patient hits go to the supplement, labelled as such

ONE-LINE SUMMARY

**CellChat turns expression into
communication hypotheses worth testing —
superb at generating them, insufficient as proof.**

If you can state the inference boundary, this episode did its job.

Check yourself 1

What is a CellChat communication score, at its core?

- A Observed cell-contact frequency under a microscope
- B An interaction probability inferred from ligand $\times$ receptor expression
- C A protein-level binding measurement
- D The physical distance between two cell types

Answer B. The score comes from database $\times$ per-type expression via mass action — no protein, secretion or spatial measurement anywhere. It is inference, not observation, which is why the verb is always 'suggests'.

Check yourself 2

A permutation p below 0.01 means what?

- A The communication is biologically strong
- B The ligand was indeed secreted
- C The score beats the label-shuffled random baseline
- D The two types must be in physical contact

Answer C. The permutation test only asks whether the score reflects the type structure — not strength, secretion or contact. Strength lives in the probability itself; report the two together.

Check yourself 3

Eight vs fourteen clusters give different top axes. What now?

- A Report whichever granularity matches expectations
- B Conclude communication analysis is unreliable and drop it
- C Inspect both, explain the gap via receptor specificity, and state the granularity in the paper
- D Average the two results

Answer C. The granularity gap is information, not a bug: it says receptor expression concentrates in a subset. Check markers and receptor expression, pick the biologically faithful level, and state the choice. Cherry-picking (A) is cheating.

NEXT EPISODE

B15: Trajectory analysis and pseudotime

Communication asks who talks to whom at one moment. Next comes time: how does a monocyte walk, step by step, into becoming a macrophage — and does your data even support drawing that axis?